

MOA-OFF: ADAPTIVE HETEROGENEOUS MODALITY-AWARE OFFLOADING WITH EDGE-CLOUD COLLABORATION FOR EFFICIENT MULTIMODAL LLM INFERENCE

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ABSTRACT

Multimodal large language models (MLLMs) enable powerful cross-modal inference but impose significant computational and latency burdens, posing severe challenges for deployment in resource-constrained environments. In this paper, we propose MoA-Off, an adaptive heterogeneous modality-aware offloading framework with edge-cloud collaboration for efficient MLLM inference. MoA-Off introduces a lightweight heterogeneous modality-aware module that estimates the complexity of heterogeneous inputs through multi-dimensional feature analysis. Then, an adaptive edge-cloud collaborative offloading strategy is proposed that dynamically schedules workloads between edge and cloud based on modality-aware complexity scores and real-time system states. The experimental results demonstrate that MoA-Off can achieve over 30% reduction in latency and 30%–65% decrease in resource overhead while maintaining competitive accuracy compared to traditional approaches.

Index Terms— Multimodal LLM, edge-cloud collaboration, adaptive offloading, inference optimization

1. INTRODUCTION

Multimodal large language models (MLLMs) have recently demonstrated remarkable capabilities in integrating visual, auditory, and textual information, enabling unified reasoning across heterogeneous data sources [1]. as shown in Fig. 1. These models are increasingly deployed in applications ranging from intelligent assistants to real-time perception systems [2]. However, their impressive performance comes at the cost of substantial computational and memory demands [3], which pose severe challenges in latency-sensitive and resource-constrained environments [4]. Pure cloud-based solutions, while powerful, often suffer from excessive transmission latency, bandwidth overhead, and potential privacy risks [5]. In contrast, edge-only approaches provide lower response time and better locality but are fundamentally limited by restricted computing capacity[6], leading to degraded performance on complex multimodal tasks [7].

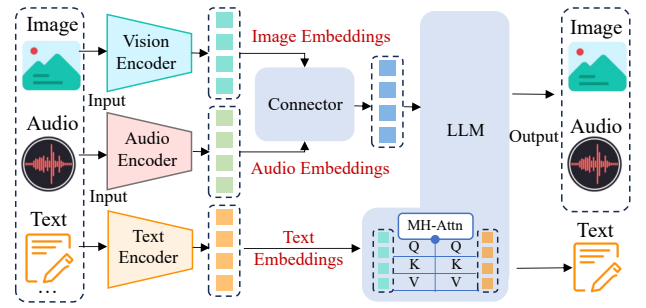


Fig. 1. The examples of MLLM.

To address these limitations, recent research has explored edge–cloud collaborative inference [8, 9], in which lightweight pre-processing is conducted on edge devices and computation-intensive reasoning is offloaded to the cloud. Such hybrid solutions balance efficiency and accuracy by distributing workloads across heterogeneous resources. Nevertheless, existing collaborative frameworks typically adopt uniform offloading policies, treating multimodal inputs without considering their intrinsic heterogeneity [11]. In practice, different modalities exhibit distinct computational characteristics, communication costs, and sensitivity to inference accuracy [12]. Ignoring these differences often results in sub-optimal allocation of tasks, inefficient resource utilization, and inconsistent user experience [13].

In this paper, we present MoA-Off, an adaptive modality-aware offloading framework for efficient MLLM inference with edge–cloud collaboration. Our approach introduces modality-specific complexity into the offloading process, thereby improving both system efficiency and inference performance. The main contributions are as follows:

- We develop a lightweight heterogeneous modality-aware module, which estimates the complexity of image and text inputs through efficient feature extraction, while maintaining minimal overhead on edge devices.
- We propose an adaptive edge-cloud collaborative of-

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抽象的

多模态大语言模型 (MLLM) 可以实现强大的跨模态推理, 但会带来巨大的计算和延迟负担, 这给资源受限环境中的部署带来了严峻的挑战。在本文中, 我们提出了 MoA-Off, 这是一种自适应异构模态感知卸载框架, 具有边云协作功能, 可实现高效的 MLLM 推理。MoA-Off 引入了一种轻量级异构模态感知模块, 该模块通过多维特征分析来估计异构输入的复杂性。然后, 提出了一种自适应边缘-云协作卸载策略, 该策略根据模态感知的复杂度分数和实时系统状态动态调度边缘和云之间的工作负载。实验结果表明, 与传统方法相比, MoA-Off 可以实现延迟降低 30% 以上, 资源开销降低 30%–65%, 同时保持有竞争力的准确性。

Index Terms— 多模态大语言模型、边云协作、自适应卸载、推理优化

1. 简介

多模态大语言模型 (MLLM) 最近在集成视觉、听觉和文本信息方面表现出了卓越的能力, 从而实现了跨异构数据源的统一推理[1]。如图 1 所示。这些模型越来越多地部署在从智能助手到实时感知系统等应用中 [2]。然而, 它们令人印象深刻的性能是以大量计算和内存需求为代价的[3], 这在延迟敏感和资源受限的环境中提出了严峻的挑战[4]。纯基于云的解决方案虽然功能强大, 但通常会遭受过大的传输延迟、带宽开销和潜在的隐私风险[5]。相比之下, 仅边缘方法提供更短的响应时间和更好的局部性, 但从根本上受到计算能力有限的限制[6], 导致复杂多模态任务的性能下降[7]。

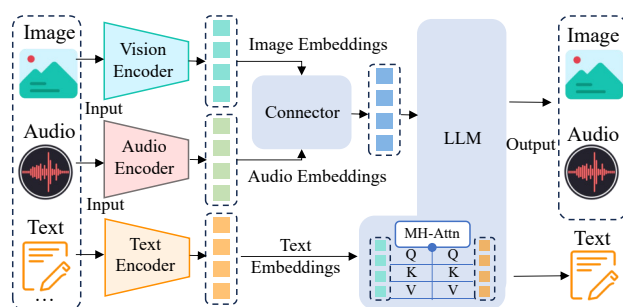


图 1. MLLM 的示例。

为了解决这些限制, 最近的研究探索了边缘-云协作推理[8, 9], 其中在边缘设备上进行轻量级预处理, 并将计算密集型推理卸载到云端。此类混合解决方案通过跨异构资源分配工作负载来平衡效率和准确性。然而, 现有的协作框架通常采用统一的卸载策略, 处理多模式输入而不考虑其内在的异质性[11]。在实践中, 不同的模态表现出不同的计算特征、通信成本和对推理准确性的敏感性[12]。忽略这些差异通常会导致任务分配不理想、资源利用效率低下以及用户体验不一致[13]。

在本文中, 我们提出了 MoA-Off, 这是一种自适应模态感知卸载框架, 用于通过边云协作进行高效 MLLM 推理。我们的方法将特定于模态的复杂性引入卸载过程, 从而提高系统效率和推理性能。主要贡献如下:

- 我们开发了一个轻量级的异构模态感知模块, 它通过有效的特征提取来估计图像和文本输入的复杂性, 同时保持边缘设备的最小开销。
- 我们提出了一种自适应边缘云协作-

flooding policy. This policy enables fine-grained scheduling, ensuring that resource-intensive modalities are offloaded to the cloud for accurate reasoning, while lightweight inputs are retained on the edge for low-latency processing.

- We conduct extensive experiments on multimodal benchmark datasets. The experimental results demonstrate that MoA-Off achieves over 30% reduction in latency while maintaining competitive accuracy, alongside a 30%–65% decrease in resource overhead, compared to traditional approaches.

2. PRELIMINARIES

Multimodal LLM. MLLM extends conventional text-based LLMs by integrating heterogeneous modalities such as text, images, audio, and video into a unified reasoning framework [14]. Formally, given an input set of modalities x_t, x_v, x_a corresponding to text, vision, and audio, the model constructs modality embeddings $h_m = f_m(x_m)$ through respective encoders $f_m(\cdot)$, and then aligns them into a common latent space via a projection function $g(\cdot)$:

$$z_m = g(h_m), \quad m \in \{t, v, a\}. \quad (1)$$

This formulation empowers MLLMs to handle complex tasks such as multimodal question answering, captioning, and grounded reasoning, yet also results in significant computational and memory demands, making efficient deployment a non-trivial challenge.

Edge-Cloud Architecture. Edge-cloud architectures provide a hierarchical computing paradigm that distributes inference workloads across resource-constrained edge devices and powerful cloud servers, thereby balancing latency, energy efficiency, and computational capacity [?]. This division enables low-latency responses while leveraging the scalability of the cloud for complex reasoning [15]. Such a collaborative design mitigates the limitations of standalone edge inference while reducing cloud dependency compared to purely centralized solutions, making it especially suitable for real-time multimodal applications under dynamic network and resource conditions.

3. THE PROPOSED METHOD

We propose MoA-Off, an adaptive heterogeneous modality-aware offloading framework designed to enhance the efficiency of MLLM inference by edge–cloud collaboration, as illustrated in Fig. 2. It primarily consists of two parts: (1) *Lightweight Heterogeneous Modality-Aware*, and (2) *Adaptive Edge-Cloud Collaborative Offloading*. The core idea of MoA-Off is to exploit the heterogeneous characteristics of different modalities and dynamically allocate computational workloads between edge and cloud.

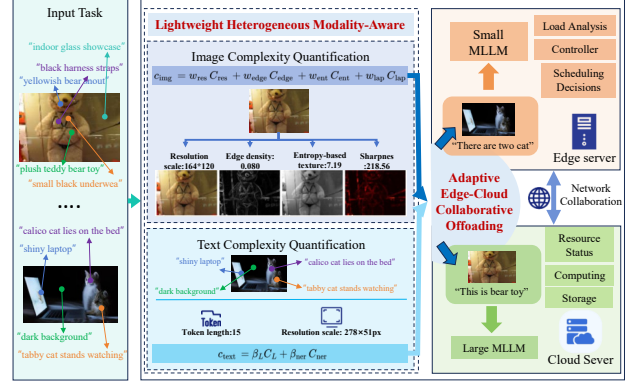


Fig. 2. The overview of the proposed MoA-Off framework.

3.1. Lightweight Heterogeneous Modality-Aware

The lightweight heterogeneous modality-aware module is designed to efficiently identify the type and estimate the complexity of each input modality on the edge device. Its primary objective is to provide lightweight yet reliable complexity scores, which serve as the basis for the adaptive offloading policy. To ensure practicality, all metrics are derived from fast, single-pass feature extraction techniques [16], making them suitable for real-time deployment in resource-constrained environments.

3.1.1. Image Complexity

For an input image $I \in [0, 255]^{H \times W}$ with $N = HW$ pixels, the image complexity is quantified as a weighted combination of several lightweight indicators that capture different aspects of visual information. Specifically, we consider resolution scale, edge density, entropy-based texture, and sharpness. The overall complexity is computed as a weighted sum: $c_{\text{img}} = w_{\text{res}}C_{\text{res}} + w_{\text{edge}}C_{\text{edge}} + w_{\text{ent}}C_{\text{ent}} + w_{\text{lap}}C_{\text{lap}}$, where the weights w are non-negative and normalized to sum to one. Each indicator is defined as follows.

Resolution scale. Higher-resolution images typically require more computation. We normalize image size relative to a reference resolution (H_0, W_0) : $C_{\text{res}} = \min\left(1, \frac{HW}{H_0W_0}\right)$.

Edge density. Images with dense edges usually contain richer structural information and thus higher processing costs. We measure the average Sobel gradient magnitude and normalize it:

$$C_{\text{edge}} = \text{clip}\left(\frac{\bar{G} - P_5(G)}{P_{95}(G) - P_5(G) + \epsilon}, 0, 1\right). \quad (2)$$

Entropy-based texture. Texture richness is approximated by image entropy, derived from the gray-level histogram:

$$H(I) = -\sum_{k=0}^{255} p_k \log p_k, \quad C_{\text{ent}} = \frac{H(I)}{\log 256}, \quad (3)$$

卸货政策。该策略可实现细粒度调度，确保将资源密集型模式卸载到云端以进行准确推理，同时将轻量级输入保留在边缘以进行低延迟处理。

- 我们对多模式基准数据集进行了广泛的实验。实验结果表明，与传统方法相比，MoA-Off 在保持有竞争力的准确性的同时，延迟时间减少了 30% 以上，同时资源开销减少了 30%–65%。

2. 准备工作

多模态大语言模型。MLLM 通过将文本、图像、音频和视频等异构模式集成到统一的推理框架中来扩展传统的基于文本的 LLM [14]。形式上，给定对应于文本、视觉和音频的模式输入集 x_t, x_v, x_a ，模型通过相应的编码器 $f_m(\cdot)$ 构造模态嵌入 $h_m = f_m(x_m)$ ，然后通过投影函数 $g(\cdot)$ 将它们对齐到公共潜在空间：

$$z_m = g(h_m), \quad m \in \{t, v, a\}. \quad (1)$$

这种表述使 MLLM 能够处理复杂的任务，例如多模态问答、字幕和基础推理，但也会导致大量的计算和内存需求，使高效部署成为一个不小的挑战。

边缘云架构。边缘云架构提供了一种分层计算范式，可以在资源受限的边缘设备和强大的云服务器之间分配推理工作负载，从而平衡延迟、能源效率和计算能力[?]。这种划分可以实现低延迟响应，同时利用云的可扩展性进行复杂推理[15]。与纯粹的集中式解决方案相比，这种协作设计减轻了独立边缘推理的局限性，同时减少了对云的依赖，使其特别适合动态网络和资源条件下的实时多模态应用。

3. 所提出的方法

我们提出了 MoA-Off，这是一种自适应异构模态感知卸载框架，旨在通过边云协作来提高 MLLM 推理的效率，如图 2 所示。它主要由两部分组成：(1)

Lightweight Heterogeneous Modality-Aware 和 (2) *Adaptive Edge-Cloud Collaborative Offloading*。MoA-Off 的核心思想是利用不同模态的异构特征，在边缘和云之间动态分配计算工作负载。

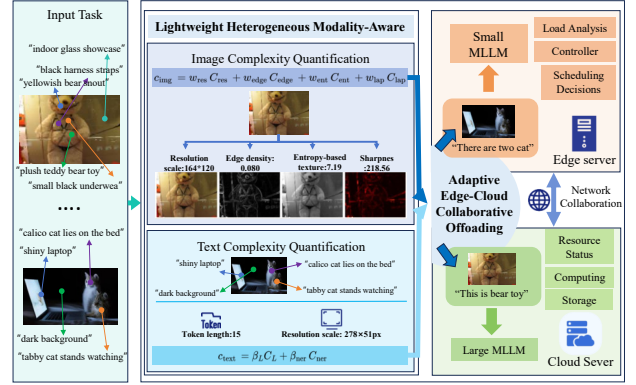


图 2. 拟议的 MoA-Off 框架概述。

3.1. 轻量级异构模态感知

轻量级异构模态感知模块旨在有效识别边缘设备上每种输入模态的类型并估计其复杂性。其主要目标是提供轻量级但可靠的复杂性分数，作为自适应卸载策略的基础。为了确保实用性，所有指标均源自快速、单通道特征提取技术[16]，使其适合在资源有限的环境中实时部署。

3.1.1. Image Complexity

对于具有 $N = HW$ 像素的输入图像 $I \in [0, 255]^{H \times W}$ ，图像复杂性被量化为捕获视觉信息不同方面的几个轻量级指标的加权组合。具体来说，我们考虑分辨率尺度、边缘密度、基于熵的纹理和清晰度。总体复杂度计算为加权和：

$C_{img} = w_{res}C_{res} + w_{edge}C_{edge} + w_{ent}C_{ent} + w_{lap}C_{lap}$ ，其中权重 w 为非负值并归一化为总和为 1。各指标定义如下。

分辨率比例。更高分辨率的图像通常需要更多的计算。我们相对于参考分辨率 (H_0, W_0) 标准化图像大小：

$$C_{res} = \min \left(1, \frac{HW}{H_0W_0} \right).$$

边缘密度。边缘密集的图像通常包含更丰富的结构信息，因此处理成本更高。我们测量平均索贝尔梯度大小并将其标准化：

$$C_{edge} = \text{clip} \left(\frac{\bar{G} - P_5(G)}{P_{95}(G) - P_5(G) + \epsilon}, 0, 1 \right). \quad (2)$$

基于熵的纹理。纹理丰富度通过从灰度直方图导出的图像熵来近似：

$$H(I) = - \sum_{k=0}^{255} p_k \log p_k, \quad C_{ent} = \frac{H(I)}{\log 256}, \quad (3)$$

where p_k is the normalized frequency of gray-level k .

Sharpness. Sharp or detailed images are more demanding to process. We use the variance of the Laplacian operator as a sharpness indicator:

$$C_{\text{lap}} = \text{clip} \left(\frac{\text{Var}(\nabla^2 I) - P_5}{P_{95} - P_5 + \epsilon}, 0, 1 \right), \quad (4)$$

where P_5 and P_{95} represent the 5-th and 95-th percentiles of the Laplacian variance across a calibration set.

3.1.2. Text Complexity

Text complexity reflects the processing overhead introduced by long sequences, entity density, and context dependency [17]. The final text complexity score is computed as: $c_{\text{text}} = \beta_L C_L + \beta_{\text{ner}} C_{\text{ner}}$, where β_L and β_{ner} are weights.

Token length. Longer sequences lead to quadratic growth in transformer computation. We normalize token count L relative to a threshold L_0 : $C_L = \min \left(1, \frac{L}{L_0} \right)$.

Entity or numeric density. The presence of named entities, numerical expressions, or specialized terminology often increases semantic reasoning complexity. We measure this factor by computing the average number of entities per sentence: $C_{\text{ner}} = \min \left(1, \frac{E/S}{\gamma} \right)$, where E is the number of entities and S is the number of sentences, γ is a scaling constant.

3.2. Adaptive Edge-Cloud Collaborative Offloading

Building upon the modality-specific complexity estimation introduced in *Section 3.1*, we design an adaptive offloading strategy that dynamically determines whether each modality should be processed locally at the edge or offloaded to the cloud. The key objective is to minimize latency and resource overhead while maintaining inference accuracy.

For each modality $m_i \in \text{text, image}$, let $c_i \in [0, 1]$ denote the estimated complexity score from the modality perception module. The system state is represented by $s = (\ell, b)$, where ℓ is the current edge load and b is the available network bandwidth. The offloading decision d_i is defined as:

$$d_i = \begin{cases} \text{edge}, & c_i \leq \tau_{m_i} \wedge \ell \leq \ell_{\max} \wedge b \leq \beta, \\ \text{cloud}, & \text{otherwise,} \end{cases} \quad (5)$$

where τ_{m_i} is the modality-specific complexity threshold, $\ell_{\max}$ is the maximum tolerable edge utilization, and β is a bandwidth limit.

In multimodal tasks, different modalities often exhibit divergent levels of complexity and resource demand. Instead of enforcing a uniform offloading decision for the entire input, our strategy employs a per-modality routing mechanism that enables partial offloading and fine-grained scheduling. For instance, in an image and text query, the image, characterized by high computational and bandwidth requirements, can be

offloaded to the cloud, while the accompanying short text is processed locally on the edge for immediate response. Formally, the decision vector is defined as

$$d = \pi(c_1, c_2, \dots, c_k, s) \in \text{edge, cloud}^k, \quad (6)$$

where k denotes the number of modalities in the input, c_i is the complexity score of modality i , s represents the system state (including edge load and network bandwidth), and $\pi(\cdot)$ is the adaptive scheduling policy function that integrates modality-aware thresholds with system-level dynamics.

4. EXPERIMENTS

4.1. Experimental Setup

We utilize an NVIDIA A100 (40 GB) GPU as the cloud server and an NVIDIA 3090 (24 GB) as the edge device. Qwen2-VL-2B model is deployed on the edge, while Qwen2.5-VL-7B model is deployed in the cloud [18]. The evaluation datasets are VQAv2 [19] and MMBench [20], we randomly selected 5,000 images from the validation set for testing. The network bandwidth is set to 200 Mbps, 300 Mbps, and 400 Mbps, respectively. The weights for image complexity and text complexity are set to their average values, with the complexity threshold set to 0.5. The baseline methods include (1) *Cloud-only* (Qwen-2.5-VL-7B), (2) *Edge-only* (Qwen2-VL-2B), and (3) *PerLLM* (an edge-cloud collaboration solution). Evaluation metrics include accuracy, End-to-end latency, and resource overhead.

4.2. Main Results and Analysis

4.2.1. Accuracy Comparison

As shown in Table 1, the proposed MoA-Off framework achieves highly competitive performance. Critically, it attains an overall accuracy that is comparable to the cloud-only approach with a negligible performance penalty of merely <0.4%. Furthermore, our method substantially outperforms the edge-only and PerLLM by a considerable margin of over 4.8%-16.8% in absolute accuracy. These results highlight the effectiveness of the proposed MoA-Off framework.

Table 1. Accuracy (%) comparison results.

Method	Cloud-only	Edge-only	PerLLM	MoA-Off
<i>VQAv2 Dataset</i>				
200 (Mbps)	76.3	61.4	71.3	76.1
300 (Mbps)	77.4	63.2	71.8	77.2
400 (Mbps)	77.8	63.5	72.4	77.5
<i>MMBench Dataset</i>				
200 (Mbps)	75.6	58.4	68.3	75.2
300 (Mbps)	76.1	60.1	69.2	75.9
400 (Mbps)	76.5	61.2	69.9	76.3

其中 p_k 是灰度级 k 的归一化频率。

锐度。清晰或详细的图像处理起来要求更高。我们使用拉普拉斯算子的方差作为锐度指标：

$$C_{\text{lap}} = \text{clip} \left(\frac{\text{Var}(\nabla^2 I) - P_5}{P_{95} - P_5 + \epsilon}, 0, 1 \right), \quad (4)$$

其中 P_5 和 P_{95} 表示校准集中拉普拉斯方差的第 5 个和第 95 个百分位数。

3.1.2. Text Complexity

文本复杂性反映了长序列、实体密度和上下文依赖性引入的处理开销[17]。最终文本复杂性得分计算如下：

$C_{\text{text}} = \beta_L C_L + \beta_{\text{ner}} C_{\text{ner}}$ ，其中 β_L 和 β_{ner} 是权重。

令牌长度。较长的序列会导致变压器计算量呈二次方增长。我们相对于阈值 L_0 标准化标记计数 L ： $C_L = \min \left(1, \frac{L}{L_0} \right)$ 。

实体或数字密度。命名实体、数字表达式或专门术语的存在通常会增加语义推理的复杂性。我们通过计算每个句子的平均实体数量来衡量这个因素： $C_{\text{ner}} = \min \left(1, \frac{E/S}{\gamma} \right)$ ，其中 E 是实体数量， S 是句子数量， γ 是缩放常数。

3.2. 自适应边缘-云协作卸载

基于 Section 3.1 中引入的特定于模式的复杂性估计，我们设计了一种自适应卸载策略，该策略动态确定每种模式是否应在边缘本地处理或卸载到云端。主要目标是最大限度地减少延迟和资源开销，同时保持推理准确性。

对于每个模式 $m_i \in \text{text}$, 图像, 令 $c_i \in [0, 1]$ 表示模式感知模块的估计复杂度得分。系统状态由 $s = (\ell, b)$ 表示，其中 ℓ 是当前边缘负载， b 是可用网络带宽。卸载决策 d_i 定义为：

$$d_i = \begin{cases} \text{edge}, & c_i \leq \tau_{m_i} \wedge \ell \leq \ell_{\max} \wedge b \leq \beta, \\ \text{cloud}, & \text{otherwise}, \end{cases} \quad (5)$$

其中 τ_{m_i} 是特定于模式的复杂性阈值， $\ell_{\max}$ 是最大可容忍边缘利用率， β 是带宽限制。

在多模式任务中，不同的模式通常表现出不同程度的复杂性和资源需求。我们的策略不是对整个输入强制执行统一的卸载决策，而是采用按模式的路由机制来实现部分卸载和细粒度调度。例如，在图像和文本查询中，具有高计算和带宽要求的图像可以被

卸载到云端，同时在边缘本地处理随附的短文本以立即响应。形式上，决策向量定义为

$$d = \pi(c_1, c_2, \dots, c_k, s) \in \text{edge, cloud}^k, \quad (6)$$

其中 k 表示输入中的模式数量， c_i 是模式 i 的复杂性得分， s 表示系统状态（包括边缘负载和网络带宽）， $\pi(\cdot)$ 是将模式感知阈值与系统级动态集成的自适应调度策略函数。

4. 实验

4.1. 实验装置

我们使用 NVIDIA A100 (40 GB) GPU 作为云服务器，使用 NVIDIA 3090 (24 GB) 作为边缘设备。Qwen2-VL-2B 模型部署在边缘，而 Qwen-2.5-VL-7B 模型部署在云端[18]。评估数据集是 VQAv2[19] 和 MMBench[20]，我们从验证集中随机选择 5000 张图像进行测试。网络带宽分别设置为 200 Mbps、300 Mbps 和 400 Mbps。图像复杂度和文本复杂度的权重设置为它们的平均值，复杂度阈值设置为 0.5。基准方法包括 (1) *Cloud-only* (Qwen-2.5-VL-7B)、(2) *Edge-only* (Qwen2-VL-2B) 和 (3) *PerLLM* (边云协作解决方案)。评估指标包括准确性、端到端延迟和资源开销。

4.2. 主要结果与分析

4.2.1. Accuracy Comparison

如表 1 所示，所提出的 MoA-Off 框架实现了极具竞争力的性能。至关重要的是，它获得了与纯云方法相当的整体准确性，并且性能损失仅为 <0.4%，可以忽略不计。此外，我们的方法在绝对精度方面大大优于仅边缘和 PerLLM，超过 4.8%-16.8%。这些结果凸显了拟议的 MoA-Off 框架的有效性。

表 1. 准确度 (%) 比较结果。

Method	Cloud-only	Edge-only	PerLLM	MoA-Off
<i>VQAv2 Dataset</i>				
200 (Mbps)	76.3	61.4	71.3	76.1
300 (Mbps)	77.4	63.2	71.8	77.2
400 (Mbps)	77.8	63.5	72.4	77.5
<i>MMBench Dataset</i>				
200 (Mbps)	75.6	58.4	68.3	75.2
300 (Mbps)	76.1	60.1	69.2	75.9
400 (Mbps)	76.5	61.2	69.9	76.3

4.2.2. End-to-End Latency Comparison

As shown in Fig. 3, the comprehensive latency measurements reveal that our proposed MoA-Off framework achieves the lowest mean end-to-end latency among all evaluated strategies. It significantly outperforms the cloud-only approach and the edge-only approach by a margin of over 50%. Compared to other edge-cloud collaborative methods, MoA-Off achieves over 30% reduction in latency. Overall, by intelligently offloading only the most complex inferences to the cloud, our method avoids the severe latency tail typical of edge-only models struggling with difficult samples, thereby achieving an optimal balance.

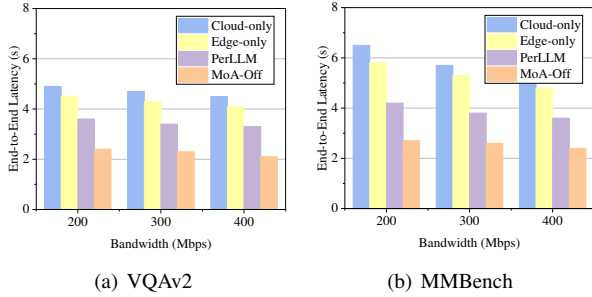


Fig. 3. The end-to-end latency comparison results of different methods. (a) shows results on the VQAv2 dataset, while (b) shows results on the MMBench dataset.

4.2.3. Resource Overhead Comparison

Computing Overhead. As shown in Fig. 4(a) and Fig. 4(b), MoA-Off exhibits the lowest computing overhead on both the VQAv2 dataset and the MMBench dataset. It drastically reduces the computational burden on the cloud infrastructure by offloading only the most complex samples. Compared to the cloud-only solution and PerLLM, MoA-Off can reduce computing overhead by 30%-65%. Our method introduces a negligible computing overhead on the edge device for calculating the modality-aware offloading policy, which is orders of magnitude lighter than running the MLMM. This strategic distribution of workload ensures that the vast majority of requests are resolved with minimal compute investment, either on the edge for simple tasks or on the cloud for complex ones.

Memory Overhead. As shown in Fig. 4(c) and Fig. 4(d), by introducing adaptive modality-aware offloading, MoA-Off minimizes unnecessary data duplication and memory retention, effectively balancing allocation between edge and cloud. The cloud-only solution, benefiting from abundant resources, incurs substantial memory costs due to centralized multi-modal model execution and frequent context reloading. In contrast, PerLLM alleviates part of this burden by distributing inference across the hierarchy, but its lack of modality-aware scheduling leads to redundant memory consumption at both

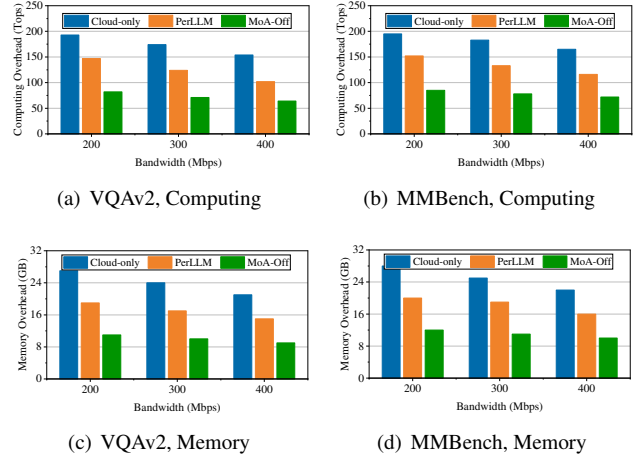


Fig. 4. The resource overhead comparison results of different methods. (a) and (b) represent computing overhead, while (c) and (d) represent memory overhead.

ends, especially when handling heterogeneous modalities with varying complexity.

4.3. Ablation Study

To further validate the effectiveness of our design, we performed ablation studies by removing the adaptive modality-aware offloading and edge-cloud collaborative scheduling components. Without the modality-aware offloading mechanism, inference accuracy dropped by 6.8% on average, reflecting the system's inability to account for heterogeneous modality complexity. Similarly, disabling the collaborative scheduling strategy increased end-to-end latency by 21.5%, while computational and memory overhead rose by 18.7% and 16.3%, respectively, due to inefficient task distribution across edge and cloud. These results clearly demonstrate that both adaptive modality-aware offloading and collaborative scheduling are indispensable for efficient MLLM inference under resource-constrained edge-cloud environments.

5. CONCLUSION

In this work, we present MoA-Off, an adaptive heterogeneous modality-aware offloading framework for efficient MLLM inference. By introducing a lightweight modality perception module and an adaptive offloading strategy, MoA-Off effectively captures the heterogeneous characteristics of different modalities and dynamically allocates workloads between edge and cloud resources. Experimental results demonstrated that the proposed framework achieves significant reductions in latency and resource overhead while maintaining competitive inference accuracy, outperforming conventional edge-only, cloud-only, and edge-cloud offloading methods.

4.2.2. End-to-End Latency Comparison

如图 3 所示，综合延迟测量表明，我们提出的 MoA-Off 框架在所有评估的策略中实现了最低的平均端到端延迟。它的性能明显优于纯云方法和纯边缘方法超过 50%。与其他边云协作方法相比，MoA-Off 的延迟降低了 30% 以上。总的来说，通过智能地将最复杂的推理卸载到云端，我们的方法避免了仅边缘模型在处理困难样本时典型的严重延迟尾部，从而实现了最佳平衡。

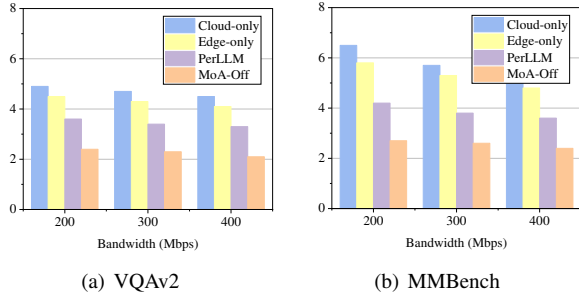


图3.不同方法的端到端延迟比较结果。(a) 显示了 VQAv2 数据集上的结果，而 (b) 显示了 MMBench 数据集上的结果。

4.2.3. Resource Overhead Comparison

计算开销。如图 4 (a) 和图 4 (b) 所示，MoA-Off 在 VQAv2 数据集和 MMBench 数据集上都表现出最低的计算开销。它仅卸载最复杂的样本，从而大大减少了云基础设施的计算负担。与纯云解决方案和 PerLLM 相比，MoA-Off 可以减少 30%-65% 的计算开销。我们的方法在计算模态感知卸载策略时在边缘设备上引入的计算开销可以忽略不计，这比运行 MLMM 轻几个数量级。这种工作负载的战略分配可确保以最少的计算投资解决绝大多数请求，无论是在边缘执行简单任务，还是在云上执行复杂任务。

内存开销。如图 4 (c) 和图 4 (d) 所示，通过引入自适应模态感知卸载，MoA-Off 最大限度地减少了不必要的数据重复和内存保留，有效平衡了边缘和云之间的分配。仅云解决方案受益于丰富的资源，但由于集中式多模式模型执行和频繁的上下文重新加载而产生大量内存成本。相比之下，PerLLM 通过在层次结构中分配推理来减轻部分负担，但其缺乏模态感知调度会导致两个方面的冗余内存消耗

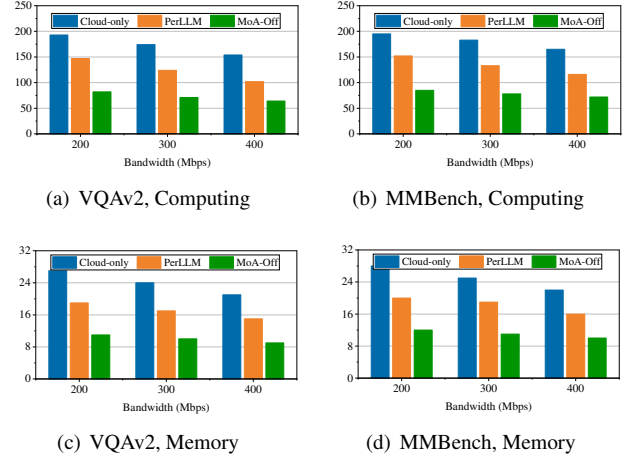


图4.不同方法的资源开销比较结果。(a)和(b)代表计算开销，而(c)和(d)代表内存开销。

结束，特别是在处理具有不同复杂性的异构模式时。

4.3. 消融研究

为了进一步验证我们设计的有效性，我们通过删除自适应模态感知卸载和边缘云协作调度组件来进行消融研究。如果没有模态感知卸载机制，推理精度平均下降 6.8%，反映出系统无法考虑异构模态复杂性。同样，禁用协作调度策略会使端到端延迟增加 21.5%，而由于边缘和云端的任务分配效率低下，计算和内存开销分别增加 18.7% 和 16.3%。这些结果清楚地表明，自适应模态感知卸载和协作调度对于资源受限的边缘云环境下的高效 MLLM 推理是不可或缺的。

5. 结论

在这项工作中，我们提出了 MoA-Off，一种用于高效 MLLM 推理的自适应异构模态感知卸载框架。通过引入轻量级模态感知模块和自适应卸载策略，MoA-Off 有效捕获不同模态的异构特征，并在边缘和云资源之间动态分配工作负载。实验结果表明，所提出的框架在保持有竞争力的推理精度的同时，显著降低了延迟和资源开销，优于传统的仅边缘、仅云和边缘云卸载方法。

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